

Loss Aversion Shifts the Coordination Boundary in Cooperative MARL: A Phase Diagram Analysis in a Multi-Echelon Supply Chain

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Abstract

Cooperative multi-agent reinforcement learning (MARL) systems fail when reward centralization falls below a critical threshold: agents converge to a zero-order *lazy-agent* policy, abandoning cooperation entirely. Despite widespread recognition of this failure mode, the collapse boundary has not been characterized as a quantitative, measurable object. The threshold is not a fixed constant—it shifts with training horizon (metastability) and, as we show, with agent-level prospect-theoretic loss aversion (λ), making it insufficient to study reward structure or behavioral bias in isolation. We construct the first quantitative phase diagram of cooperative MARL coordination collapse in (reward centralization w , training horizon T , loss aversion λ) space, using pre-registered collapse definitions and bootstrap-estimated critical points with confidence intervals. Across 1,124 controlled experiments on a multi-echelon supply chain substrate, we identify $w_{\text{crit}} = 0.666$ [0.635, 0.694] at 300k steps under no behavioral bias. A three-way null-model decomposition reveals that probability weighting alone ($\alpha=\beta=0.88$, $\lambda=1$) shifts w_{crit} *left* by 0.100 [0.033, 0.172]—facilitating coordination—while loss aversion at $\lambda=3$ and $\lambda=7$ shifts it *right* by 0.089 [0.041, 0.141] and 0.166 [0.135, 0.202] respectively. Probability weighting and loss aversion exert statistically significant effects in *opposite directions*, a decomposition that, to our knowledge, has not previously been demonstrated in cooperative MARL.

1 Introduction

Automated replenishment in multi-echelon supply chains is a canonical cooperative MARL problem: agents must coordinate ordering decisions across nodes with delayed feedback, partial observability, and misaligned local incentives (Oroojlooyjadid et al., 2022). Failures in such systems—agents that under-order, hoard, or stop ordering altogether—generate the bullwhip effect, stockouts, and cascading profit losses. Deploying MARL agents that exhibit human-like behavioral biases, such as prospect-theoretic loss aversion, is increasingly realistic given the trend toward human-in-the-loop and behavior-aware AI systems (Lalmohammed et al., 2025; Ethayarajh et al., 2024).

A well-documented failure mode is *coordination collapse*: agents converge to a zero-action policy, abandoning cooperation entirely (Liu et al., 2023; Shi and Peng, 2023). While the phenomenon is recognized, no prior work has characterized the collapse boundary as a quantitative, measurable object with confidence intervals. Existing work either treats the boundary qualitatively (“centralization helps”) or studies algorithmic mitigation without locating the transition point.

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The boundary is not a fixed scalar. It depends on the reward structure (w , the fraction of global reward shared across agents), the training horizon T , and—as we demonstrate—on whether agents carry prospect-theoretic loss aversion (λ). These factors interact: measuring any one in isolation does not characterize the system. Prospect theory decomposes into two components with distinct mathematical forms (Tversky and Kahneman, 1992): a nonlinear probability weighting function (α, β) and a loss aversion coefficient (λ). Prior work applying prospect theory to cooperative MARL (Lalmohammed et al., 2025) integrates these components jointly into the algorithm, making it impossible to attribute effects to either component separately.

We close this gap by constructing the first quantitative phase diagram of cooperative MARL coordination collapse. We treat the collapse boundary as a logistic curve in w -space, estimate the critical point w_{crit} with bootstrap confidence intervals, and use a pre-registered null-model design to decompose the total behavioral effect into a *probability weighting component* (curvature) and a *loss aversion increment* (λ -dependent).

Our main results are:

- The structural coordination boundary is $w_{\text{crit}} = 0.666$ [0.635, 0.694] at 300k steps under no behavioral bias, and $w_{\text{crit}} = 0.822$ [0.785, 0.858] at 150k steps, demonstrating that the boundary is horizon-dependent (metastable attractor).
- Probability weighting alone ($\alpha=\beta=0.88$, $\lambda=1$) *reduces* w_{crit} by 0.100 [0.033, 0.172] relative to the no-PT null, facilitating coordination.
- Loss aversion at $\lambda=3$ and $\lambda=7$ shifts w_{crit} by +0.089 [0.041, 0.141] and +0.166 [0.135, 0.202] above the null, respectively. The non-overlapping confidence intervals of the $\lambda=7$ and $\lambda=1$ conditions confirm the behavioral claim.
- The two PT components act in *opposite directions*: curvature facilitates coordination; loss aversion impedes it.

Figure 1 shows the collapse probability surface in (w, T) space, illustrating the metastability result. Figure 2 shows the behavioral decomposition: the four logistic curves separate clearly, confirming both the curvature and loss-aversion effects.

2 Background

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (Lowe et al., 2017) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local ($w=0$) and purely global ($w=1$) reward.

When $w \approx 0$, each agent’s reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate’s action improves the global outcome. Liu et al. (2023) and Shi and Peng (2023) formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (Kahneman and Tversky, 1979) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative

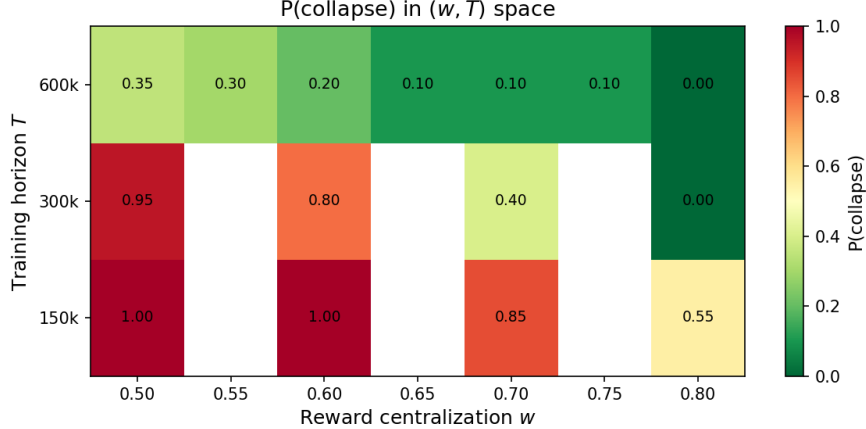


Figure 1: $P(\text{collapse})$ in (w, T) space. Each cell shows the empirical collapse fraction across 20 seeds. The 150k and 300k rows are from Exp A ($\alpha=\beta=1$); the 600k row is from Exp T ($\alpha=\beta=0.88$). Because $\alpha=\beta=0.88$ shifts w_{crit} left by 0.100 (§6.3), the apparent 300k→600k slide is partially curvature-inflated. Under matched $\alpha=\beta=1$ conditions (Exp 0b), $P(\text{collapse} \mid w=0.6) = 0.35$ and $P(\text{collapse} \mid w=0.7) = 0.20$ at 600k, confirming that metastability holds with $\alpha=\beta$ held constant; only the magnitude is curvature-inflated in this figure.

prospect theory (CPT) (Tversky and Kahneman, 1992) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^\alpha & \text{if } x \geq 0 \\ -\lambda(-x)^\beta & \text{if } x < 0, \end{cases} \quad (1)$$

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting $\alpha=\beta=1$ and $\lambda=1$ recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for agent i is:

$$r_i^{\text{PT}} = v\left((1-w)r_i^{\text{local}} + w r^{\text{global}}\right), \quad (2)$$

where $v(\cdot)$ is given by Equation (1). This differs architecturally from Lalmohammed et al. (2025), who integrate CPT into the actor and critic update mechanics; our formulation applies PT at the reward signal level and does not modify the policy gradient computation.

3 Related Work

The lazy agent problem (Liu et al., 2023) and relative overgeneralization (Shi and Peng, 2023) establish coordination collapse as a canonical MARL pathology under CTDE, proposing algorithmic mitigations (CMAE and CURO respectively). Weng and Lee (2026) demonstrate that exploration noise structurally degrades shared neural representations under decentralised training, explaining a key mechanism of collapse. Akella (2025) show empirically that even sophisticated decentralised reward shaping cannot overcome fundamental coordination barriers, achieving -24.20 average reward

versus a centralised baseline at +1.92. These works establish *why* collapse occurs; none characterises *where* the collapse boundary lies as a quantitative object with confidence intervals. Zhao and Xie (2025) construct counterfactual advantage functions at multiple credit-assignment levels and evaluate robustness across thresholding hyperparameters—a sweep that borders on a centralization sensitivity analysis—but they optimise win rates on SMAC rather than deriving a statistical phase boundary in (w, T) space. Jiang et al. (2020) and Zhang et al. (2026) similarly utilise reward mixing weights and observe phase-like transitions in training dynamics, respectively, but neither isolates a named critical centralization value with confidence intervals.

Most recently, Yamaguchi (2025) construct a phase diagram for independent Q-learning across environment size L and agent density ρ , identifying coordinated, fragile, and jammed regimes separated by an instability ridge. Our work is orthogonal: we hold the environment and algorithm fixed and map the phase boundary as a function of the reward structure parameter w and the behavioral parameter λ —dimensions their framework does not address.

Lalmohammed et al. (2025) integrate CPT into the actor and critic update mechanics of MADDPG, demonstrating that tunable risk profiles are achievable in cooperative settings. Our work is architecturally distinct (CPT as reward transform, not gradient transform) and asks a different question: not whether PT-equipped agents can coordinate, but at what centralization level coordination fails and how the PT parameter λ shifts that level.

Qiu et al. (2021) and Shen et al. (2023) incorporate Conditional Value at Risk (CVaR) into cooperative value factorisation. CVaR optimises the expected return in the worst α -percentile of outcomes; loss aversion operates via reference-dependent utility asymmetry. These are mathematically distinct mechanisms: CVaR addresses aleatoric tail risk, while λ captures the perceived asymmetry between gains and losses (Shen et al., 2023).

Mengesha et al. (2025) prove in evolutionary game theory that loss aversion induces population-level lock-in at zero-sum equilibria, establishing the conceptual phenomenon in a theoretical setting. We extend this to deep MARL with learned neural policies and continuous state spaces, providing the first empirical measurement in this setting.

Oroojlooyjadid et al. (2022) establish the Beer Distribution Game as a multi-agent Dec-POMDP where decentralised DQN triggers systemic cost escalation corrected by a shared penalty. Gijsbrechts et al. (2022) demonstrate that A3C can match state-of-the-art heuristics in multi-echelon inventory settings, establishing DRL as a viable approach for supply chain optimisation. Leluc et al. (2023) apply MARL to multi-product single-echelon inventory. Our substrate generalises these to a six-node multi-echelon network.

4 Method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\text{collapsed}(\text{run}) = \mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1], \quad (3)$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit 03c3fc6). Section A demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\text{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\text{collapsed} \mid w) = \sigma(b_0 + b_1 w), \quad (4)$$

Table 1: Experiment summary. All experiments use the same environment and algorithm. $\alpha=\beta=1$ denotes the linear (no-PT) condition.

Exp	Purpose	w values	λ	α, β	Seeds	Runs
A @150k	Structural boundary	$\{0.0, 0.1, \dots, 1.0\}$	1	1,1	20	220
A @300k	Structural boundary	$\{0.2, \dots, 0.8\}$	1	1,1	20	140
C	Basin control	$\{0.5, 0.85, 1.0\}$	1	1,1	12	144
N	Null baseline	$\{0.0, 0.2, \dots, 1.0\}$	1	1,1	20	180
B-Red.	Behavioral surface	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{1, 3, 7\}$	0.88	20	300
T (600k only)	Metastability horizon	$\{0.5, 0.55, \dots, 0.80\}$	1	0.88	20	140

where σ is the logistic function. We estimate $w_{\text{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. We report Wilson score 95% confidence intervals for per- w collapse fractions.

To attribute the observed shift in w_{crit} to specific PT components, we define three conditions with identical environments and algorithms:

1. **Null (Exp N):** $\alpha=\beta=1$, $\lambda=1$ —the linear, no-PT condition.
2. **Curvature (Exp B, $\lambda=1$):** $\alpha=\beta=0.88$, $\lambda=1$ —probability weighting with no loss aversion.
3. **Loss aversion (Exp B, $\lambda \in \{3, 7\}$):** $\alpha=\beta=0.88$, $\lambda > 1$ —full PT.

The curvature effect is defined as $w_{\text{crit}}(\text{B}, \lambda=1) - w_{\text{crit}}(\text{N})$, and the loss aversion increment as $w_{\text{crit}}(\text{B}, \lambda) - w_{\text{crit}}(\text{B}, \lambda=1)$. This decomposition is valid because the null removes all PT distortion, while the curvature condition adds nonlinearity without loss asymmetry.

5 Experimental Setup

We use **NetworkManagement-v1** from **or-gym** (Hubbs et al., 2020): a six-node multi-echelon inventory network with Poisson demand ($\mu=20$) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that maps each agent’s continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position.

Each agent runs PPO (Schulman et al., 2017) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (Raffin et al., 2021), PettingZoo (Terry et al., 2021), and SuperSuit (Terry et al., 2020) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

Table 1 summarises all six experiments. All designs were pre-registered before execution. Total run count: 1,124.

6 Results

6.1 Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions

rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation.

The critical boundary decreases monotonically with training horizon T : $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, $\alpha=\beta=1$). Figure 1 visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

Note on the 600k row. The 600k row in Figure 1 comes from Exp T, which uses $\alpha=\beta=0.88$. Because our decomposition shows $\alpha=\beta=0.88$ shifts w_{crit} left by 0.100 relative to $\alpha=\beta=1$, the 600k collapse fractions in the figure are partially curvature-deflated, not purely a horizon effect. To bound the horizon-only slide, we use Exp 0b ($\alpha=\beta=1$, 20 seeds, 600k): $P(\text{collapse} \mid w=0.6) = 0.35$ and $P(\text{collapse} \mid w=0.7) = 0.20$, compared to 0.80 and 0.40 at 300k. The metastability result—the boundary slides left with T —holds under matched $\alpha=\beta=1$ conditions; only the magnitude at 600k in the figure is inflated by the curvature effect.

Exp C rules out initialization as a driver of collapse. Within- w correlations between S_{init} and converged S are -0.11 ($w=0.5$), -0.38 ($w=0.85$), and -0.36 ($w=1.0$)—all near zero and inconsistent with initialization dominating the outcome. The collapsed/survived assignment is not sensitive to where agents start.

6.2 Null Model and Reproducibility

Exp N (null: $\alpha=\beta=1$, $\lambda=1$) yields $w_{\text{crit}} = 0.685$ [0.657, 0.710], compared to $w_{\text{crit}} = 0.666$ [0.635, 0.694] for the PT ($\lambda=1$, $\alpha=\beta=1$) condition from Exp A @300k—a difference of -0.020 with overlapping confidence intervals. Both conditions use no loss aversion and linear value functions. Their agreement to within 0.020 confirms that the w_{crit} measurement reproduces across independent experimental runs, validating the null baseline.

6.3 Behavioral Phase Boundary

Exp B-Reduced yields monotonically increasing w_{crit} with loss aversion λ : $w_{\text{crit}}(\lambda=1) = 0.586$ [0.539, 0.624], $w_{\text{crit}}(\lambda=3) = 0.774$ [0.751, 0.798], and $w_{\text{crit}}(\lambda=7) = 0.851$ [0.845, 0.858].¹ The behavioral claim is confirmed: $w_{\text{crit}}(\lambda=7)$ lower CI 0.845 exceeds $w_{\text{crit}}(\lambda=1)$ upper CI 0.624—the confidence intervals are non-overlapping by 0.221. Figure 3 illustrates the monotone relationship.

Table 2 and Figure 2 present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** ($\alpha=\beta=0.88$, $\lambda=1$): excess = -0.100 , 95% CI [$-0.172, -0.033$]. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.
2. **Loss aversion at $\lambda=3$** : excess = $+0.089$, 95% CI [$+0.041, +0.141$] (CI excludes zero; significant).

¹For $\lambda=7$, the near-complete separation of outcomes (collapse fractions 1.00, 1.00, 1.00, 0.55, 0.00 at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$) renders the logistic slope and p-value unreliable. We report w_{crit} from bootstrap estimation, which is robust to quasi-separation; the slope and p-value are not reported for this condition.

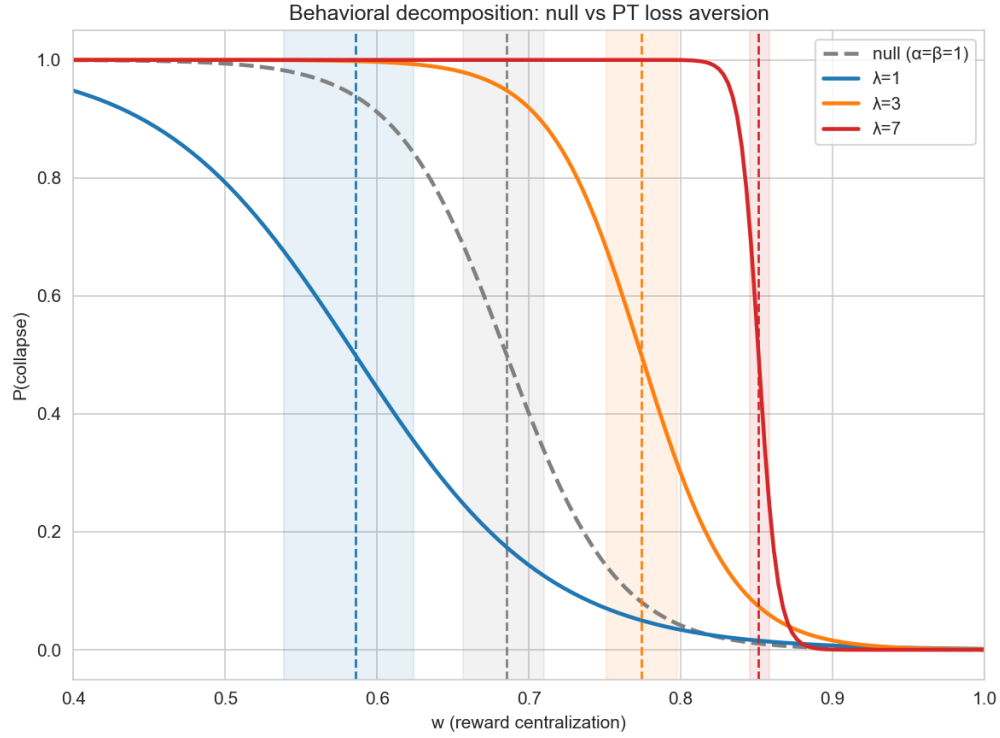


Figure 2: Behavioral decomposition of the collapse boundary. Each curve shows $P(\text{collapse} \mid w)$ at 300k steps. The null ($\alpha=\beta=1$, gray dashed) is the structural baseline. The $\lambda=1$, $\alpha=\beta=0.88$ curve (blue) shifts *left* of the null by 0.100, indicating that probability weighting alone facilitates coordination. The $\lambda=3$ (orange) and $\lambda=7$ (red) curves shift progressively *right*, indicating that loss aversion raises the centralization requirement. Vertical dashed lines mark w_{crit} estimates; shaded bands show bootstrap 95% confidence intervals.

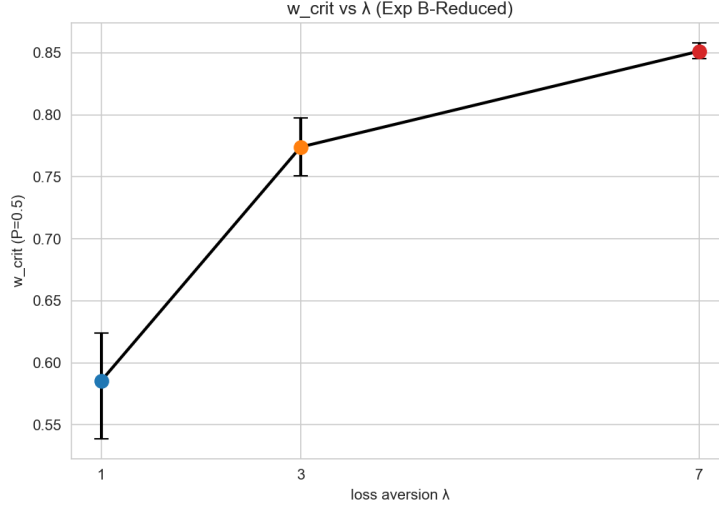


Figure 3: w_{crit} as a function of loss aversion λ (Exp B-Reduced, $\alpha=\beta=0.88$, 300k steps, 20 seeds per cell). Error bars show bootstrap 95% confidence intervals. The monotone increase and non-overlapping CIs between $\lambda=1$ and $\lambda=7$ confirm the behavioral claim. Note: the $\lambda=7$ CI [0.845, 0.858] should be interpreted with caution due to quasi-complete separation; see Section 6.3.

Table 2: Three-way behavioral decomposition of w_{crit} relative to the null model. “Significant” denotes that the conservative difference confidence interval excludes zero. [†]For $\lambda=7$, w_{crit} is from bootstrap only; see footnote in Section 6.3.

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null ($\alpha=\beta=1$, $\lambda=1$)	0.685	[0.657, 0.710]	0	—	—
$\alpha=\beta=0.88$, $\lambda=1$	0.586	[0.539, 0.624]	−0.100	[−0.172, −0.033]	✓ (neg.)
$\alpha=\beta=0.88$, $\lambda=3$	0.774	[0.751, 0.798]	+0.089	[+0.041, +0.141]	✓
$\alpha=\beta=0.88$, $\lambda=7^{\dagger}$	0.851	[0.845, 0.858]	+0.166	[+0.135, +0.202]	✓

3. **Loss aversion at $\lambda=7$:** excess = +0.166, 95% CI [+0.135, +0.202] (CI excludes zero; significant).

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At $\lambda=7$, the loss aversion increment above the curvature baseline is +0.265 ($= 0.851 - 0.586$), overwhelming the curvature benefit of −0.100 for a net shift of +0.166 above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

7 Discussion

The finding that probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel. Prior work applies prospect theory as a unified behavioural model without decomposing its components (Lalmohammed et al., 2025; Ghaemi et al., 2024). Our null-model design isolates the two components, revealing that the beneficial effect of curvature (−0.100) is masked when combined with loss aversion ($\lambda>1$), which reverses the sign and amplifies the required centralization. This has a concrete operational interpretation: a system with $\lambda=7$ agents—calibrated to reflect human loss aversion estimates (Tversky and Kahneman, 1992)—requires

$w > 0.851$ for reliable coordination, compared to $w > 0.666$ for risk-neutral agents. The additional 16.6 percentage points of centralization attributable to behavioral bias ($w_{\text{crit}}(\lambda=7) - w_{\text{crit}}(\text{null}) = 0.851 - 0.685 = 0.166$, consistent with the excess in Table 2) are a quantified cost of deploying loss-averse agents in a cooperative system.

The sliding w_{crit} result is a positive finding, not a pathology. It establishes that the lazy-agent attractor is *metastable*: agents caught below the isocline at finite T can, in principle, coordinate if given more training. This is consistent with Weng and Lee (2026)’s finding that exploration breaks cooperation in shared-policy MARL during early training, before representations converge. The practical implication is that the minimum w required depends on the operational training budget: tighter horizons demand higher centralization.

Yamaguchi (2025) construct a phase diagram in (L, ρ) space for IQL, identifying analogous coordinated/fragile/jammed regimes. Our results complement theirs by demonstrating that the phase structure is not only a function of environmental complexity but also of the reward structure (w) and agent behavioral parameters (λ). Together, these works suggest that MARL coordination exhibits coherent phase structure across multiple parameter axes.

Several limitations should be noted. First, all results are demonstrated on a single environment (NetworkManagement-v1) and algorithm (PPO); replication across environments and algorithms remains future work. Second, the $\lambda=7$ condition exhibits quasi-complete separation; additional w values above 0.85 would sharpen the w_{crit} estimate. Third, the metastability result describes finite-horizon behavior; an asymptotic lower bound on w_{crit} is not estimated by our experiments. Fourth, we fix the demand distribution and lead times; the dependence of w_{crit} on these environmental parameters is unexplored.

8 Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. A three-way null-model decomposition reveals that the two components of prospect theory—probability weighting and loss aversion—exert statistically significant effects in opposite directions on the coordination boundary: the former facilitates coordination by 0.100; the latter impedes it by up to 0.166. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across algorithms (MAPPO, QMIX) and environments (MPE, SMAC), a mean-field theoretical derivation of the logistic boundary shape, adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous- λ populations.

A Threshold Robustness

We verify that the collapse definition (Equation (3)) is robust to perturbation of its thresholds. We re-estimate w_{crit} on Exp A @300k data under nine variants of the definition, varying $\bar{S}$ threshold over $\{8, 10, 12\}$ and $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with threshold $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1 , so no run satisfies this condition); they are excluded. Across the remaining six estimable variants, the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the confidence interval width. We conclude that the collapse definition is robust to $\pm 20\%$ threshold perturbation.

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